

Saurabh Joshi, PhD

Director of Engineering & Applied Researcher

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PROFESSIONAL SUMMARY

Engineering executive and applied researcher (PhD, Computer Science) with a track record of taking novel ideas from mathematical design to production-grade, mission-critical systems. Proven ability to build and scale multi-team engineering organizations in resource-constrained startup environments while maintaining high engineering rigor. Combines deep expertise in distributed systems architecture, software reliability, and compiler-level optimization with hands-on product leadership — having conceptualized and shipped platforms adopted across 80+ partner networks. Experienced in establishing security-first engineering cultures, driving quality through automated testing and formal verification, and serving as a trusted technical voice to executive leadership, partners, and developer communities.

CORE COMPETENCIES

Engineering Leadership

- Multi-Team Scaling & Org Design
- Talent Acquisition & Mentorship
- Strategic Roadmapping & Delivery
- Cross-Functional Collaboration
- Stakeholder & Partner Engagement

Systems Architecture

- Distributed Systems & Consensus
- Low-Latency Data Pipelines
- Event-Driven Execution Platforms
- VM / Language Runtime Design
- Database Schema Optimization

Software Trust & Security

- Formal Verification & Specification
- Automated Testing & Coverage
- Security Audits & Incident Response
- DevSecOps & CI/CD Governance
- IP Generation (Patents & Research)

TECHNICAL EXPERTISE

Languages: C/C++, Move, Rust, Python, Solidity · **Verification & Solvers:** SAT/MaxSAT (Open-WBO), Symbolic Execution (Pinaka, CBMC), Model Checking, Formal Specification · **Compiler Infrastructure:** LLVM, Static Analysis, Program Transformation · **Platforms & VMs:** MoveVM, EVM · **Data & Infrastructure:** PostgreSQL, MySQL, GCP, GitHub Actions, CI/CD Pipelines

PROFESSIONAL EXPERIENCE

Consulting Advisor *Cerify.ai*

Jul 2026–Present Remote

- **Strategic Advisory:** Advising leadership on technological roadmap in the security and audit space, shaping product direction for automated vulnerability detection.
- **Engineering Mentorship:** Mentoring the entire engineering team on architecting solutions for automated vulnerability detection engines, bringing formal methods and static analysis expertise to core detection capabilities.

Director of Engineering and Innovations *Supra*

Oct 2024–Present Remote

- **Organizational Leadership:** Lead 4 product engineering teams (Smart Contracts, Execution Layer, Oracle, Automation — 14+ engineers) and coordinate cross-functionally with Layer 1, Wallet, DeFi, and Explorer teams to deliver integrated platform releases.
- **Product Conceptualization (0 → 1):** Conceived and drove the creation of a native event-driven automation platform, eliminating dependency on external off-chain services. Leveraged existing VM execution paths to accelerate delivery, achieving zero-block-delay execution guarantees and scaling to 50,000 concurrent automation tasks.
- **High-Performance Data Platform:** Steered the architecture of the core real-time data distribution protocol (DORA), delivering sub-second latency for 600+ data feeds per second, adopted across 80+ partner networks.
- **Quality & Engineering Culture:** Instituted code coverage enforcement, PR review checklists and templates, and championed a culture of unit testing, property-based testing, and modular, reusable design across all teams. Pioneered AI-assisted test generation to maintain velocity under resource constraints.
- **Infrastructure Optimization:** Recommended PostgreSQL schema modifications that reduced database storage requirements by 20%, directly lowering operational costs.

- **Security & Incident Response:** Led incident response for multiple critical security events — analyzing blast radius, coordinating cross-team fixes, and executing expedited production releases. Established 24/7 operational coverage through DevOps on-call rotations, emergency escalation chains, and automated liveness monitoring with alerting.
- **Governance & Partner Relations:** Served as a trusted member of the multi-signature governing authority managing protocol-level security and upgrades. Engaged directly with partners and enterprise clients to define integration requirements and drive adoption.
- **Executive Communication:** Reported directly to the CEO on technical strategy, requirements, and crisis response. Represented the company at developer events and authored technical papers and blog posts.

Principal Engineer *Supra*

May 2022–Sep 2024 Remote

- **Engineering Rescue & Turnaround:** Assumed technical ownership of the DORA oracle protocol after the departure of the previous engineering director. Inherited a partial, incorrect, and non-modular implementation; redesigned the architecture for correctness, modularity, and cross-network compatibility.
- **Novel Systems Design:** Invented a merkle-tree chunking scheme enabling a single cryptographic signature over 600+ data feeds to be distributed to networks with heterogeneous payload size constraints — reducing bandwidth and verification costs. Optimized provenance data flow by offloading responsibility to data-gatherer nodes, cutting storage and network overhead.
- **Consistent Cross-Platform APIs:** Designed and shipped a unified API and client SDK layer consumed across multiple runtime environments, ensuring a consistent developer experience for external integrators.
- **Research to Production:** Pioneered the mathematical design of the DORA consensus protocol, securing a US Patent and publishing a peer-reviewed paper at IEEE ICDCS 2023.
- **Formal Verification Strategy:** Initiated the company's formal verification framework and contract specification strategy to mathematically prove correctness of critical execution paths.

Staff Engineer, Product Security *Qualcomm*

Oct 2021–May 2022 Hyderabad, India

- **Team Leadership (14 Engineers):** Led and managed a Security Validation team of 14 engineers designing proof-of-concept exploits and security mitigations for the mobile software stack. Validation work directly secured chipsets shipped in commercial mobile devices.
- **Process Optimization (70% Faster PR Cycles):** Diagnosed a systemic code review bottleneck caused by context-switching and deprioritized reviews. Engineered an automated metrics pipeline using the GitHub API to surface developer and reviewer dwell times, sending transparent reports to the team — reducing PR cycle time by 70%.
- **Talent Pipeline:** Cultivated strategic university relations across India to build a high-caliber talent pipeline. Personally involved in hiring, interviewing, and recommending engineers who became strong contributors.
- **People Management:** Directed performance evaluations, career development, and mentorship programs for the combined engineering team.

Assistant Professor Grade I *Department of CSE, IIT Hyderabad*

May 2016–Sep 2021 Hyderabad, India

- **Applied R&D Leadership:** Headed a research group (2 PhD students, 4 Masters students, and undergraduate researchers) pursuing applied systems research in program analysis, compiler verification, software security, and constraint optimization.
- **Grant Acquisition & Budget Management:** Secured grants totaling over ₹4.9 Crore (\$600K) in competitive research funding and industry consultancies from national agencies (DST, NSCS, Ministry of Defense, CDAC) and international corporations (Suzuki Motor Corporation, Japan).
- **Industry Consultancies:**
 - *Suzuki Motor Corporation, Japan:* Consulted on optimizing build times and binary sizes for Autograde Linux systems.
 - *Ministry of Defense:* Developed MISRA C code scanners for safety-critical embedded software compliance.
 - *NSCS:* Contributed to formulating security standards for operating systems and firmware in national network infrastructure devices.
- **Award-Winning Open-Source Tools:** Developed and open-sourced Pinaka (symbolic execution engine), Open-WBO-Inc (optimization solver), and GPURepair (automated repair of concurrency bugs in GPU kernels). Tools won multiple Gold/Silver medals at international solver competitions.

Assistant Professor (Grade I) *Department of CSE, IIT Guwahati*

Sep 2015–May 2016 Guwahati, India

- Conducted research and directed graduate thesis projects in computer science and systems engineering.

Post Doctoral Research Fellow *Department of Computer Science, University of Oxford* Aug 2012–Aug 2015 Oxford, UK

- Developed formal verification frameworks (2LS, cbmc-repair, Open-WBO) under major international research initiatives, collaborating with academic and industrial partners across Europe.

Research Intern *Rigorous Software Engineering group, Microsoft Research India* **May 2010–Apr 2011** Bangalore, India

- Developed automated static analysis tools for detecting concurrency bugs and under-specified software harnesses in enterprise systems. Work resulted in a US Patent and a publication at POPL 2012.

Project Trainee *Software Engineering group, IBM India Research Lab* **Aug 2006–Jul 2007** New Delhi, India

- Contributed to enterprise-scale static analysis frameworks for automated software verification.

PATENTS

- **Distributed Oracle Agreement System and Method (DORA)** – US Patent App. PCT/US/36923
Saurabh Joshi, Aniket Kate, Joshua Tobkin and David Yang
- **Finding Bugs with Low False Alarms and Under-specified Harness** – US Patent 8793664
Saurabh Joshi, Shuvendu Lahiri and Akash Lal

PUBLICATIONS & RESEARCH

Peer-reviewed research in top-tier international venues (POPL, IJCAI, IEEE ICDCS, VMCAI, CP, ACM TOPLAS, ACM TACO) spanning distributed systems, program analysis, formal verification, constraint optimization, and software security. Full publication record: sbjoshi.github.io/publications

EDUCATION

Doctor of Philosophy, Computer Science *Indian Institute of Technology (IIT) Kanpur* **Jul 2007–Feb 2013** Kanpur, India

Thesis: Program verification, automated bug-finding, and concurrency analysis.

Master of Technology (Research Assistantship), Computer Science *Indian Institute of Technology (IIT) Bombay* **Jul 2003–Jul 2006** Mumbai, India

Bachelor of Engineering, Information Technology *GCET, Sardar Patel University* **Sep 1999–Jul 2003** Gujarat, India